

```
In [35]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: data = pd.read_csv(r"seattle-weather.csv")
data.head()
```

```
Out[2]:
```

	date	precipitation	temp_max	temp_min	wind	weather
0	2012-01-01	0.0	12.8	5.0	4.7	drizzle
1	2012-01-02	10.9	10.6	2.8	4.5	rain
2	2012-01-03	0.8	11.7	7.2	2.3	rain
3	2012-01-04	20.3	12.2	5.6	4.7	rain
4	2012-01-05	1.3	8.9	2.8	6.1	rain

```
In [3]: data.shape
```

```
Out[3]: (1461, 6)
```

```
In [4]: data.isna().sum()
```

```
Out[4]: date          0
precipitation      0
temp_max          0
temp_min          0
wind              0
weather           0
dtype: int64
```

```
In [5]: data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 1461 entries, 0 to 1460
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  -
0   date            1461 non-null  str
1   precipitation    1461 non-null  float64
2   temp_max        1461 non-null  float64
3   temp_min        1461 non-null  float64
4   wind            1461 non-null  float64
5   weather         1461 non-null  str
dtypes: float64(4), str(2)
memory usage: 88.0 KB
```

```
In [19]: # set 'date' dtype to datetime
data['date'] = pd.to_datetime(data['date'])
```

```
In [20]: data.weather.value_counts()
```

```
Out[20]: weather
rain      641
sun       640
fog       101
drizzle   53
snow      26
Name: count, dtype: int64
```

```
In [21]: data.describe()
```

Out[21]:

	date	precipitation	temp_max	temp_min	wind
count	1461	1461.000000	1461.000000	1461.000000	1461.000000
mean	2013-12-31 00:00:00	3.029432	16.439083	8.234771	3.241136
min	2012-01-01 00:00:00	0.000000	-1.600000	-7.100000	0.400000
25%	2012-12-31 00:00:00	0.000000	10.600000	4.400000	2.200000
50%	2013-12-31 00:00:00	0.000000	15.600000	8.300000	3.000000
75%	2014-12-31 00:00:00	2.800000	22.200000	12.200000	4.000000
max	2015-12-31 00:00:00	55.900000	35.600000	18.300000	9.500000
std	NaN	6.680194	7.349758	5.023004	1.437825

```
In [22]: data.duplicated().sum()
```

```
Out[22]: np.int64(0)
```

```
In [23]: df = data.sort_values(by='date').copy()
```

```
In [24]: df
```

Out[24]:

	date	precipitation	temp_max	temp_min	wind	weather
0	2012-01-01	0.0	12.8	5.0	4.7	drizzle
1	2012-01-02	10.9	10.6	2.8	4.5	rain
2	2012-01-03	0.8	11.7	7.2	2.3	rain
3	2012-01-04	20.3	12.2	5.6	4.7	rain
4	2012-01-05	1.3	8.9	2.8	6.1	rain
...	...	...	...	...	...	...
1456	2015-12-27	8.6	4.4	1.7	2.9	rain
1457	2015-12-28	1.5	5.0	1.7	1.3	rain
1458	2015-12-29	0.0	7.2	0.6	2.6	fog
1459	2015-12-30	0.0	5.6	-1.0	3.4	sun
1460	2015-12-31	0.0	5.6	-2.1	3.5	sun

1461 rows × 6 columns

```
In [25]: # set index
df.set_index('date', inplace=True)
```

```
In [26]: df.head()
```

Out[26]:

	precipitation	temp_max	temp_min	wind	weather
date					
2012-01-01	0.0	12.8	5.0	4.7	drizzle
2012-01-02	10.9	10.6	2.8	4.5	rain
2012-01-03	0.8	11.7	7.2	2.3	rain
2012-01-04	20.3	12.2	5.6	4.7	rain
2012-01-05	1.3	8.9	2.8	6.1	rain

```
In [27]: df.resample('W')['temp_max'].mean()
```

```
Out[27]: date
2012-01-01    12.800000
2012-01-08     9.285714
2012-01-15     5.457143
2012-01-22     3.728571
2012-01-29     8.414286
...
2015-12-06    10.742857
2015-12-13    10.957143
2015-12-20     7.471429
2015-12-27     5.400000
2016-01-03     5.850000
Freq: W-SUN, Name: temp_max, Length: 210, dtype: float64
```

```
In [29]: df.resample('W')['precipitation'].max()
```

```
Out[29]: date
2012-01-01     0.0
2012-01-08    20.3
2012-01-15     5.3
2012-01-22    19.8
2012-01-29    27.7
...
2015-12-06    15.7
2015-12-13    54.1
2015-12-20    21.8
2015-12-27    27.4
2016-01-03     1.5
Freq: W-SUN, Name: precipitation, Length: 210, dtype: float64
```

```
In [30]: df.resample('YE')['precipitation'].mean()
```

```
Out[30]: date
2012-12-31    3.349727
2013-12-31    2.268493
2014-12-31    3.377534
2015-12-31    3.121096
Freq: YE-DEC, Name: precipitation, dtype: float64
```

```
In [31]: df.resample('ME')[['precipitation', 'temp_max', 'temp_min', 'wind']].agg(['min', 'max',
```

Out[31]:

date	precipitation			temp_max			temp_min			w		
	min	max	mean	min	max	mean	min	max	mean	min	max	mean
2012-01-31	0.0	27.7	5.590323	-1.1	12.8	7.054839	-3.3	7.2	1.541935	1.3	8.2	3.900
2012-02-29	0.0	17.3	3.182759	5.0	16.1	9.275862	-2.2	7.8	3.203448	1.3	8.1	3.903
2012-03-31	0.0	27.4	5.903226	5.0	15.6	9.554839	-1.7	7.2	2.838710	2.1	7.0	4.248
2012-04-30	0.0	10.9	2.270000	8.9	23.3	14.873333	1.7	10.0	5.993333	1.6	8.0	3.373
2012-05-31	0.0	18.5	1.683871	11.1	26.7	17.661290	3.9	12.8	8.190323	1.3	6.4	3.354
2012-06-30	0.0	16.5	2.503333	12.8	24.4	18.693333	6.1	15.0	10.480000	1.7	6.4	3.083
2012-07-31	0.0	15.2	0.848387	18.3	28.3	22.906452	9.4	15.0	12.932258	1.7	6.0	2.877
2012-08-31	0.0	0.0	0.000000	21.1	34.4	25.858065	10.0	18.3	14.009677	1.8	4.7	2.829
2012-09-30	0.0	0.3	0.030000	16.1	32.2	22.880000	7.8	14.4	11.243333	1.1	5.6	2.646
2012-10-31	0.0	34.5	5.493548	7.8	23.9	15.829032	3.3	13.3	8.380645	1.3	7.3	3.532
2012-11-30	0.0	54.1	7.016667	7.8	17.8	11.326667	-0.6	12.8	5.226667	1.0	6.0	3.220
2012-12-31	0.0	22.6	5.612903	3.3	13.3	7.235484	-1.7	8.3	3.293548	1.7	9.5	3.832
2013-01-31	0.0	38.4	3.409677	1.1	11.7	6.106452	-4.4	7.2	0.796774	0.6	7.3	2.877
2013-02-28	0.0	9.4	1.439286	6.1	13.3	9.467857	1.1	7.2	4.325000	1.3	8.1	3.632
2013-03-31	0.0	11.9	2.248387	7.2	20.6	12.709677	0.0	11.1	4.977419	1.1	7.6	3.461
2013-04-30	0.0	39.1	4.986667	7.8	21.7	14.243333	3.3	10.6	6.696667	1.1	6.4	3.713
2013-05-31	0.0	13.7	1.951613	11.1	30.6	19.625806	3.3	13.9	9.922581	1.0	6.5	3.158
2013-06-30	0.0	9.9	1.103333	17.2	33.9	23.253333	10.0	18.3	13.163333	1.3	5.7	3.003

date	precipitation			temp_max			temp_min			w		
	min	max	mean	min	max	mean	min	max	mean	min	max	mean
2013-07-31	0.0	0.0	0.000000	19.4	31.7	26.093548	11.1	18.3	13.932258	1.4	4.6	2.687
2013-08-31	0.0	19.3	1.109677	17.2	31.1	26.119355	13.3	18.3	15.480645	1.0	4.6	2.541
2013-09-30	0.0	43.4	5.226667	13.9	33.9	21.360000	7.2	17.2	13.590000	1.4	7.1	3.013
2013-10-31	0.0	9.1	1.264516	10.0	22.8	14.229032	3.3	11.7	7.638710	0.4	6.3	2.274
2013-11-30	0.0	30.0	3.210000	7.8	17.8	12.053333	-0.5	11.7	5.590000	0.5	7.9	3.180
2013-12-31	0.0	10.7	1.367742	0.0	13.3	7.022581	-7.1	8.3	1.570968	0.5	8.8	2.735
2014-01-31	0.0	21.6	3.032258	5.6	14.4	9.600000	-0.5	10.0	4.096774	0.8	8.8	3.348
2014-02-28	0.0	26.4	5.542857	-1.6	14.4	8.200000	-6.0	7.8	2.635714	0.8	7.9	4.528
2014-03-31	0.0	46.7	7.741935	7.2	18.9	12.906452	1.1	10.6	5.425806	1.6	6.2	3.677
2014-04-30	0.0	18.5	3.536667	11.1	27.8	15.460000	4.4	9.4	6.730000	2.3	5.8	3.696
2014-05-31	0.0	33.3	2.580645	13.3	29.4	19.870968	7.2	13.3	10.216129	1.4	5.6	3.154
2014-06-30	0.0	6.4	0.626667	15.6	26.1	21.590000	8.9	14.4	11.756667	2.1	5.0	3.190
2014-07-31	0.0	19.3	0.632258	18.9	34.4	26.900000	11.7	17.8	14.425806	1.1	5.4	3.096
2014-08-31	0.0	21.6	1.483871	17.8	35.6	26.383871	11.1	17.8	14.893548	0.6	4.1	2.477
2014-09-30	0.0	20.3	1.890000	16.7	32.2	23.163333	10.0	16.1	13.233333	1.2	6.5	3.143
2014-10-31	0.0	32.0	5.532258	12.8	25.6	17.961290	6.7	13.9	10.883871	1.0	5.4	2.916
2014-11-30	0.0	34.3	4.103333	2.8	16.7	11.030000	-4.9	12.2	4.510000	0.8	7.7	3.983
2014-12-31	0.0	20.6	3.929032	3.3	18.9	10.138710	-3.2	10.6	4.609677	1.1	6.7	3.564

date	precipitation			temp_max			temp_min			wind		
	min	max	mean	min	max	mean	min	max	mean	min	max	mean
2015-01-31	0.0	26.2	3.000000	5.0	17.2	10.154839	-3.2	11.1	4.351613	0.5	6.6	2.4290
2015-02-28	0.0	26.2	4.792857	9.4	16.7	12.517857	0.6	10.0	6.085714	0.9	6.6	3.0780
2015-03-31	0.0	55.9	3.661290	10.6	20.6	14.377419	-0.5	10.6	6.193548	1.3	5.7	2.9320
2015-04-30	0.0	14.0	1.720000	11.1	25.0	15.503333	2.8	10.6	6.030000	1.7	6.5	3.3130
2015-05-31	0.0	6.1	0.477419	12.2	27.8	20.025806	6.1	12.8	10.129032	1.8	5.2	2.8120
2015-06-30	0.0	4.6	0.196667	16.1	33.3	26.063333	9.4	18.3	13.576667	2.1	5.0	3.2460
2015-07-31	0.0	2.0	0.074194	21.1	35.0	28.093548	12.2	17.8	15.500000	1.9	4.3	2.9830
2015-08-31	0.0	32.5	2.687097	18.3	33.3	26.087097	12.2	17.2	14.693548	1.7	5.8	3.1540
2015-09-30	0.0	5.8	0.703333	15.6	27.2	20.293333	7.2	15.0	11.366667	1.3	6.8	3.0500
2015-10-31	0.0	33.0	3.948387	12.2	23.3	17.538710	7.2	13.9	10.500000	1.1	7.2	3.0350
2015-11-30	0.0	47.2	7.086667	1.7	15.6	9.683333	-3.8	9.4	3.480000	0.9	8.0	3.5460
2015-12-31	0.0	54.1	9.177419	4.4	15.6	8.380645	-2.1	10.0	3.825806	1.3	7.6	4.3410

```
In [42]: df_year = df.resample('YS')[['precipitation', 'temp_max', 'temp_min', 'wind']].agg(['min', 'max', 'mean'])
df_year
```

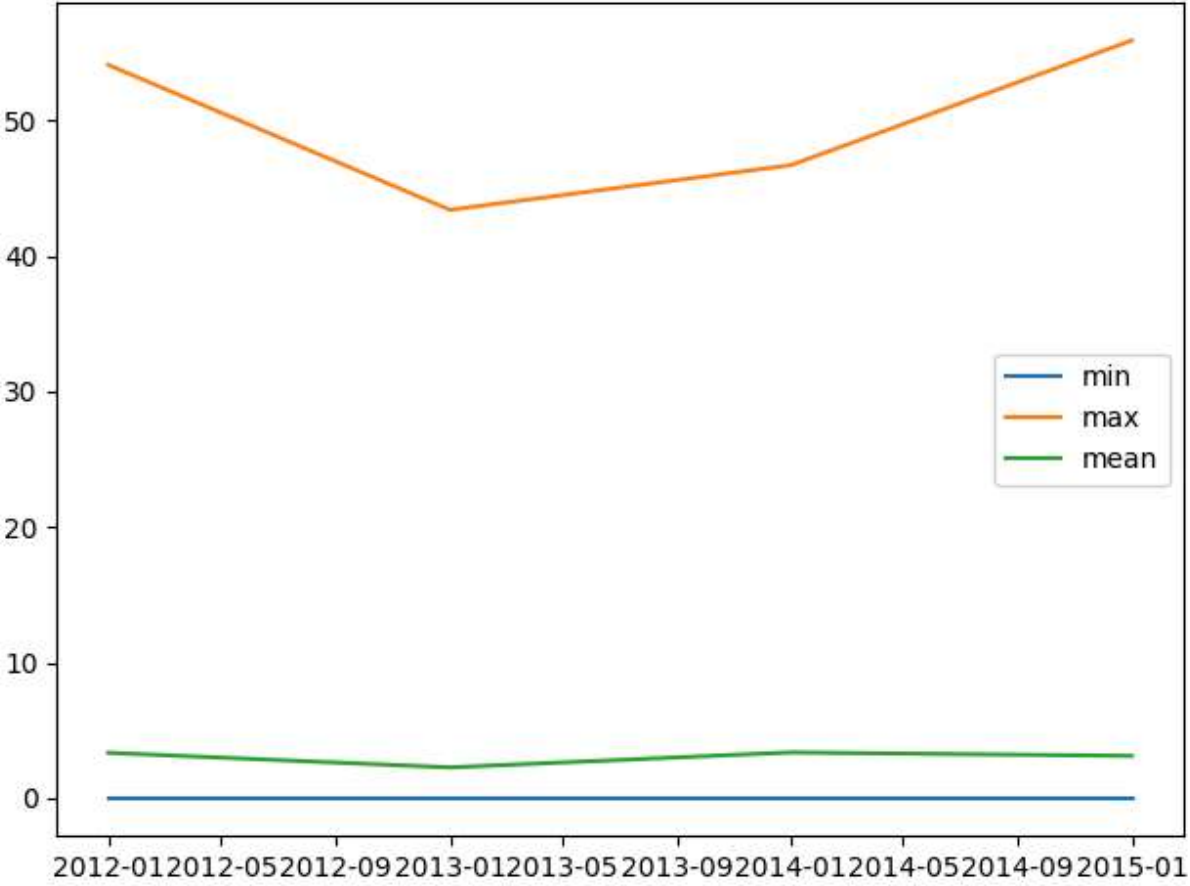
Out[42]:

	date	precipitation			temp_max			temp_min					
		min	max	mean	min	max	mean	min	max	mean	min	max	
0	2012-01-01	0.0	54.1	3.349727	-1.1	34.4	15.276776	-3.3	18.3	7.289617	1.0	9.5	3.40
1	2013-01-01	0.0	43.4	2.268493	0.0	33.9	16.058904	-7.1	18.3	8.153973	0.4	8.8	3.01
2	2014-01-01	0.0	46.7	3.377534	-1.6	35.6	16.995890	-6.0	17.8	8.662466	0.6	8.8	3.38
3	2015-01-01	0.0	55.9	3.121096	1.7	35.0	17.427945	-3.8	18.3	8.835616	0.5	8.0	3.15

In [61]:

```
plt.plot(df_year['date'], df_year['precipitation'], label=['min', 'max', 'mean'])
plt.tight_layout()
plt.legend()
```

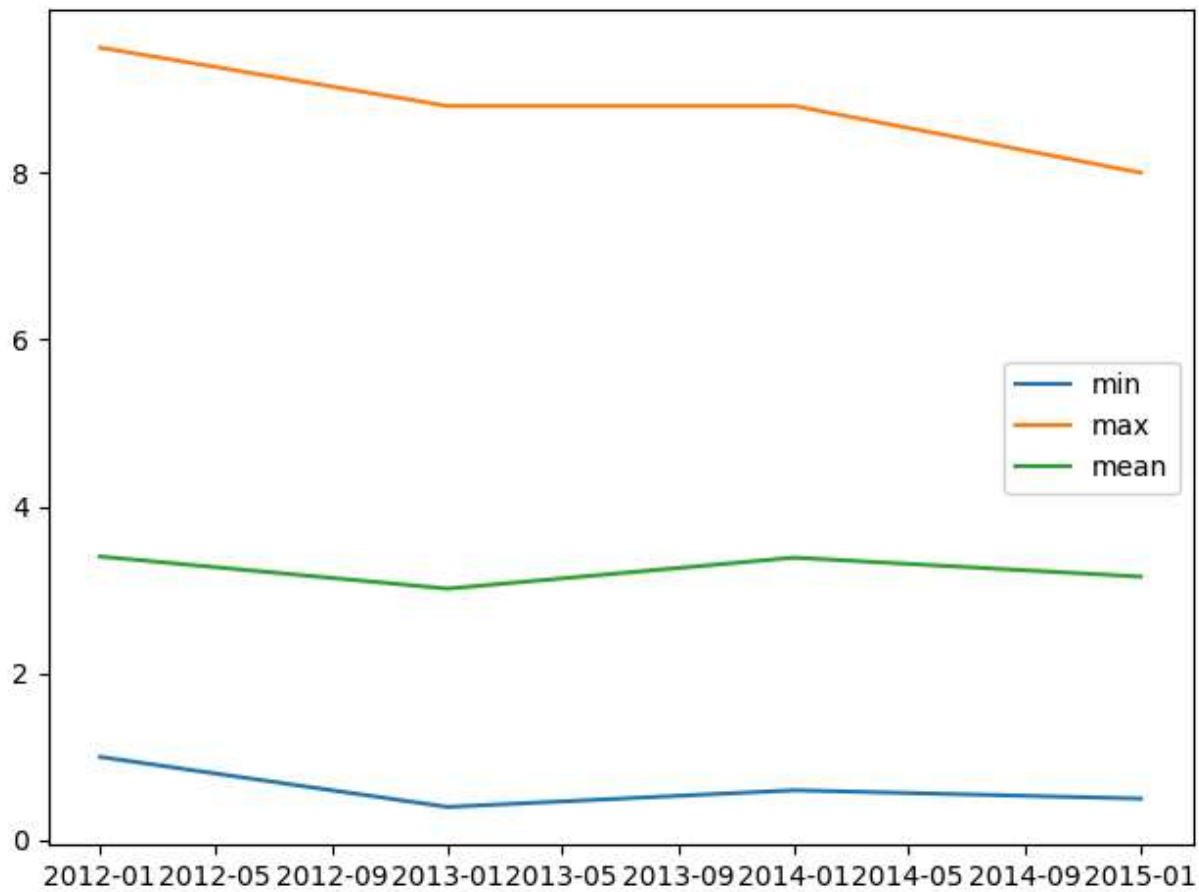
Out[61]: <matplotlib.legend.Legend at 0x20cfde3d550>



In [62]:

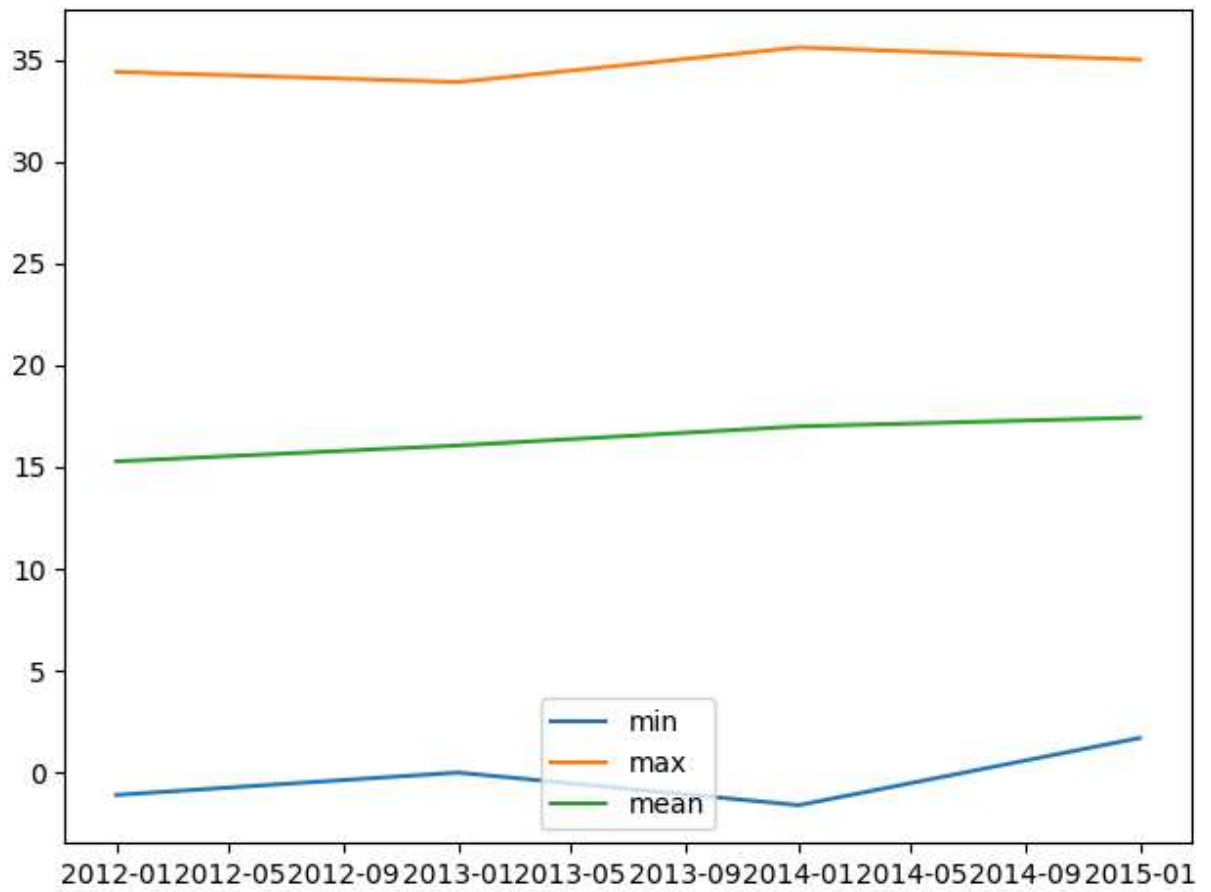
```
plt.plot(df_year['date'], df_year['wind'], label=['min', 'max', 'mean'])
plt.tight_layout()
plt.legend()
```

Out[62]: <matplotlib.legend.Legend at 0x20cfde4cec0>



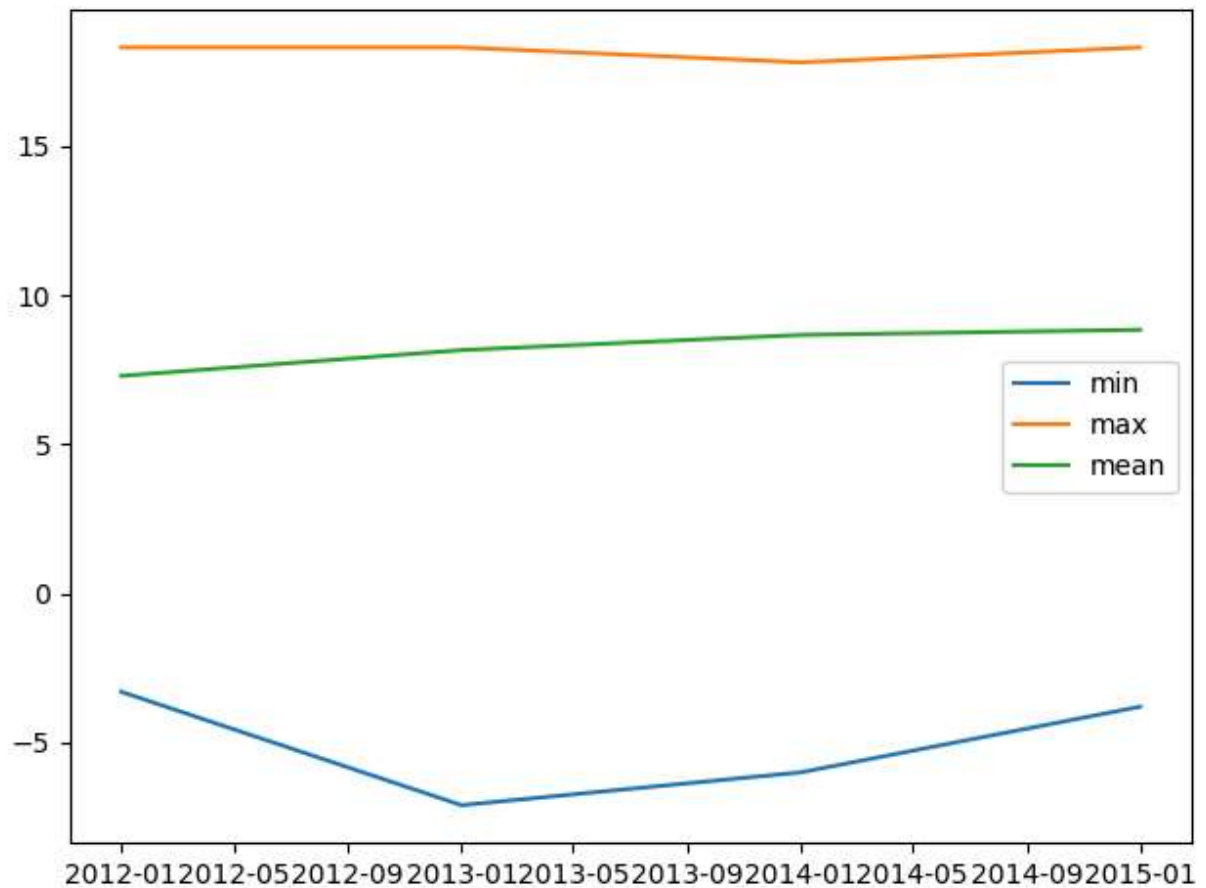
```
In [63]: plt.plot(df_year['date'], df_year['temp_max'], label=['min', 'max', 'mean'])  
plt.tight_layout()  
plt.legend()
```

```
Out[63]: <matplotlib.legend.Legend at 0x20cfe1106e0>
```



```
In [64]: plt.plot(df_year['date'], df_year['temp_min'], label=['min','max','mean'])  
plt.tight_layout()  
plt.legend()
```

```
Out[64]: <matplotlib.legend.Legend at 0x20cfe169be0>
```



```
In [65]: from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report, accuracy_score
```

```
In [66]: # X, y parameters
df.head(2)
```

```
Out[66]:      precipitation  temp_max  temp_min  wind  weather
```

date					
2012-01-01	0.0	12.8	5.0	4.7	drizzle
2012-01-02	10.9	10.6	2.8	4.5	rain

```
In [67]: X = df[['precipitation', 'temp_max', 'temp_min', 'wind']]
y = df['weather']
```

```
In [68]: X.head(2)
```

Out[68]:

	precipitation	temp_max	temp_min	wind
date				
2012-01-01	0.0	12.8	5.0	4.7
2012-01-02	10.9	10.6	2.8	4.5

In [69]: `y.head(2)`

Out[69]:

```
date
2012-01-01    drizzle
2012-01-02     rain
Name: weather, dtype: str
```

In [70]: `le = LabelEncoder()`

In [71]: `y_encoded = le.fit_transform(y)`
`y_encoded`

Out[71]: `array([0, 2, 2, ..., 1, 4, 4], shape=(1461,))`

In [73]: `df['weather_encoded'] = y_encoded`

In [75]: `y = y_encoded`
`y`

Out[75]: `array([0, 2, 2, ..., 1, 4, 4], shape=(1461,))`

In [76]: `# X,y split into train and test`
`X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2, random_state=42)`

In [77]: `# create model`
`from sklearn.ensemble import RandomForestClassifier`

In [78]: `rf = RandomForestClassifier()`

In [79]: `rf.fit(X_train,y_train)`

Out[79]:

RandomForestClassifier ⓘ ?

► Parameters

► Fitted attributes

In [81]: `y_pred = rf.predict(X_test)`
`y_pred`

```
Out[81]: array([4, 2, 4, 0, 2, 2, 4, 4, 2, 4, 4, 4, 2, 4, 4, 2, 4, 2, 4, 4, 2, 4,
                2, 4, 4, 2, 4, 4, 4, 4, 2, 4, 4, 3, 4, 2, 2, 2, 2, 4, 2, 4, 4, 4,
                4, 2, 2, 4, 2, 4, 4, 2, 4, 2, 4, 2, 2, 4, 2, 4, 2, 4, 2, 4, 4, 4,
                2, 4, 2, 2, 2, 4, 4, 2, 2, 4, 2, 4, 2, 2, 2, 4, 2, 4, 2, 4, 4, 2,
                4, 2, 2, 4, 4, 2, 2, 4, 4, 2, 4, 2, 4, 4, 2, 3, 2, 4, 4, 2, 4, 2,
                2, 4, 2, 2, 4, 0, 4, 4, 4, 2, 2, 2, 2, 2, 4, 4, 4, 4, 4, 4, 4, 4,
                4, 0, 4, 4, 2, 2, 2, 4, 4, 4, 2, 2, 4, 4, 4, 2, 4, 4, 2, 4, 4, 4,
                2, 2, 1, 4, 2, 2, 4, 4, 2, 4, 1, 2, 2, 4, 4, 4, 2, 2, 4, 4, 2, 2,
                4, 4, 4, 2, 4, 4, 4, 2, 2, 4, 2, 2, 4, 4, 4, 4, 4, 2, 4, 4, 2, 2,
                2, 2, 2, 4, 0, 2, 4, 2, 2, 4, 3, 4, 2, 4, 4, 4, 4, 4, 4, 4, 2, 4,
                2, 4, 4, 4, 4, 2, 4, 2, 4, 2, 4, 2, 4, 4, 4, 4, 4, 4, 4, 2, 2,
                4, 2, 4, 4, 4, 4, 2, 2, 2, 2, 2, 4, 4, 4, 2, 4, 2, 2, 2, 2, 4, 2,
                4, 4, 2, 1, 4, 4, 4, 1, 4, 2, 4, 4, 2, 2, 4, 4, 4, 2, 4, 2, 4, 4,
                2, 4, 4, 2, 4, 2, 4])
```

```
In [82]: acc_score = accuracy_score(y_test,y_pred)
         print("Random Forest Accuracy score is : ", acc_score)
```

Random Forest Accuracy score is : 0.825938566552901

```
In [85]: print("classification report: ")
         print(classification_report(y_test,y_pred, target_names = le.classes_))
```

classification report:

	precision	recall	f1-score	support
drizzle	0.25	0.11	0.15	9
fog	0.50	0.08	0.14	25
rain	0.93	0.93	0.93	120
snow	0.67	0.25	0.36	8
sun	0.78	0.96	0.86	131
accuracy			0.83	293
macro avg	0.62	0.47	0.49	293
weighted avg	0.80	0.83	0.79	293

```
In [98]: df.head(10)
```

Out[98]:

	precipitation	temp_max	temp_min	wind	weather	weather_encoded
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date						
2012-01-01	0.0	12.8	5.0	4.7	drizzle	0
2012-01-02	10.9	10.6	2.8	4.5	rain	2
2012-01-03	0.8	11.7	7.2	2.3	rain	2
2012-01-04	20.3	12.2	5.6	4.7	rain	2
2012-01-05	1.3	8.9	2.8	6.1	rain	2
2012-01-06	2.5	4.4	2.2	2.2	rain	2
2012-01-07	0.0	7.2	2.8	2.3	rain	2
2012-01-08	0.0	10.0	2.8	2.0	sun	4
2012-01-09	4.3	9.4	5.0	3.4	rain	2
2012-01-10	1.0	6.1	0.6	3.4	rain	2

```
In [94]: # data set
new_day = pd.DataFrame([
    "precipitation": 0.0,
    "temp_max": 16.5,
    "temp_min": 6.0,
    "wind": 3.2
])
```

In [95]: new_day

Out[95]:

	precipitation	temp_max	temp_min	wind
0	0.0	16.5	6.0	3.2

```
In [96]: pred = rf.predict(new_day)
pred
```

Out[96]: array([4])

```
In [97]: print("Pred Weather: ", le.inverse_transform(pred))
```

Pred Weather: ['sun']

```
In [99]: # new day record
new_day1 = pd.DataFrame([
    "precipitation": 0.0,
    "temp_max": 12.8,
    "temp_min": 5.0,
    "wind": 4.7
])
```

```
In [100... pred1 = rf.predict(new_day1)
print("Weather type: ", le.inverse_transform(pred1))
```

Weather type: ['drizzle']

```
In [ ]:
```